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NECESSITY AS THE MOTHER OF 'GREEN' INVENTIONS: INSTITUTIONAL PRESSURES AND ENVIRONMENTAL INNOVATIONS

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Drawing on institutional theory and innovation literature, we argue that greater regulatory and normative pressures concerning environmental issues positively influence companies' propensity to engage in environmental innovation. Analysis of environment-related patents of 326 publicly traded firms from polluting industries in the United States suggests that institutional pressures can trigger such innovation, especially in those firms displaying a greater deficiency gap (i.e., firms polluting relatively more than their industry peers). Moreover, we find that this effect is stronger when asset specificity is high, and that the availability of resources plays different roles depending on the type of pressures (regulatory vs. normative). Copyright © 2012 John Wiley & Sons, Ltd.

INTRODUCTION

In recent years, firms have been increasingly scrutinized for wrongdoing, particularly concerning the environment (such as release of toxic materials and emissions that augment global warming). Investors allegedly discount share prices of 'polluting' companies, governments introduce policies that place a cost on emissions, and consumers consider a company's environmental philosophy when purchasing products and services (Hart, 1995). One possible response to these mounting pressures is environmental innovation, that is, the development of products, processes, and services aimed at reducing environmental harm by using new methods for treating emissions, recycling or reusing waste, finding cleaner energy sources, and so on (Brunnermeier and Cohen, 2003).

Compared with other environmental practices, environmental innovation is riskier, requires greater financial commitment, and usually accrues returns in the long term (Ahuja, Lampert, and Tandon, 2008; Scherer, 1999). Moreover, unlike other environmental practices that are often off-the-shelf alternatives oriented toward meeting minimal environmental standards and that can be obtained in the open market (Berrone and Gomez-Mejia, 2009), environmental innovation emphasizes pollution prevention. It concerns the invention of new designs and the creation of novel products and processes to reduce or eliminate the use and generation of hazardous substances. Hence, environmental innovation has important

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externalities because it can lead to a cleaner and safer world. Some have argued that it might be a source of competitive advantage because it is potentially valuable to the firm's consumers and other major stakeholders (Hart, 1995; Sharma and Vredenburg, 1998).

While there is wide consensus about the value of environmental innovation, at least from a social perspective, little is known as to why some firms engage in more environmental innovation than others and, perhaps more important, under what conditions firms pursue this type of innovation. Drawing on insights from institutional theory and innovation literature, we argue that greater regulatory pressures (from governments) and normative pressures (from non-governmental organizations) make it more attractive for the focal firm to engage in environmental innovation. Moreover, not all firms facing similar regulatory and normative pressures will respond in the same fashion. We argue that firms with a greater deficiency gap (i.e., firms polluting relatively more than their peers) are an easy target of critics. Thus, underperforming firms should be more sensitive to institutional pressures and respond with greater diligence in the search for novel approaches to overcome these environmental problems. We further suggest that firms are more likely to engage in environmental innovation when they have sufficient slack in internal resources, and when they are heavily invested in firm-specific assets. We test our arguments using longitudinal data (1997–2001) on 326 publicly traded firms from polluting industries in the United States. Overall, results support our contentions.

This study makes several contributions to the current literature. First, we focus on the conditions under which firms pursue environmental innovation. The extant literature has explored the determinants of the adoption of several organizational practices that can reduce firms' impact on the natural environment. For instance, Short and Toffel (2010) looked at self-regulatory structures, Delmas and Toffel (2008) focused on both the ISO 14001 standard and non-government initiated voluntary environmental programs, and Delmas and Montes-Sancho (2010) studied firms' participation in the Climate Challenge program. By contrast, little is known about what would explain the variations across firms in terms of environmental innovation.

Second, we contribute to the empirical research on this topic by applying a novel measure to assess a firm's environmental innovation through patents. In previous studies, environmental innovation has been commonly measured through questionnaire surveys (e.g., Anton, Deltas, and Khanna, 2004; Christmann, 2000). However, this type of data could be seriously biased as respondents may tend to present a socially desirable image of themselves or their companies. While some scholars have examined environment-related patents at an aggregate country or industry level (Brunnermeier and Cohen, 2003; Jaffe and Palmer, 1997), company-level analyses are almost nonexistent.

Third, we argue and find empirical support for the notion that institutional pressures engender heterogeneous rather than isomorphic organizational responses. Variation in organizational responses to institutional pressures has been suggested theoretically (Oliver, 1991, 1997) and has been the focus of recent research explaining the adoption of environmental management practices such as ISO 14001 or the participation in voluntary governmental programs (Delmas and Toffel, 2008). But these practices can be considered as 'ceremonial conformity' responses to institutional stimuli and thus more superficial and cosmetic (Borial, 2007). Little is known, however, about more substantive responses to external demands such as coming up with inventive solutions to environmental problems. We extend current research on diverse organizational responses to these demands by arguing that similar 'green' institutional pressures lead to unique and differentiated 'green' innovation on the part of firms, and that differences in the deficiency gap engender heterogeneous responses to similar pressures.

Lastly, we bridge the institutional and resource-based view (RBV) approaches (see, for instance, Oliver, 1997) by examining organizational slack and resource specificity as moderators in how firms respond to institutional pressures on behalf of the environment. In short, our study contributes to a theoretical understanding of why some firms engage in more environmental innovation than others, analyzing institutional factors as predictors and internal firm resources as moderators.

THEORETICAL FRAMEWORK AND HYPOTHESES

Institutional theory (DiMaggio and Powell, 1983; Meyer and Rowan, 1977; Scott, 1995, 2005) focuses on how social influence toward conformity

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shapes organizations' actions. Organizations are assumed to seek approval and thus are susceptible to social influence. One of the main theses of institutionalists is that organizations try to enhance or protect their legitimacy (Deephouse, 1999; Scott, 1995). Concern over legitimacy, in turn, induces firms to adopt practices that are socially valuable within an institutional field.

Institutional theory does not delve into efficiency issues nor the impact of strategic choices on firm performance, because financial considerations are not the primary driver of socially compliant managerial practices (c.f. Berrone and Gomez-Mejia, 2009). This feature has made the theory particularly attractive to environmental management researchers, since 'green investments' often cannot be financially justified, at least in the short term (e.g., Bansal, 2005; Bansal and Clelland, 2004; Hoffman, 1999, 2000). Several studies during the past decade or so illustrate this intellectual tradition. Hoffman (1999) examined the chemical and petroleum industries from 1960 to 1993 and reported that these industries had changed in response to mounting institutional pressures for better environmental performance even though he could not uncover any financial gains as a result of these changes. Also applying institutional theory, Delmas (2002) described the variation in the implementation of ISO 14001 across Europe and the United States in ways that might not have been justified from an efficiency or economic perspective. More recently, Berrone *et al*. (2010) showed that family firms are more sensitive to environmental institutional pressures and exhibit better environmental performance, even though there is no evidence of financial benefits to the firm as a result of these environmental responses.

Most of the research noted above is driven by the assertion that, in their quest for legitimacy, firms become more alike as institutional pressures increase. This 'copycat' behavior is evidenced by the diffusion of ISO standards in the industrialized world (Delmas, 2002). Perhaps because isomorphism is so ingrained in institutionalist writings, scholars have tended to ignore the extent to which 'green' inventions may be a viable option to cope with external pressures. We argue that strong regulatory and normative institutional pressures make environmental innovation more attractive to firms, especially to those displaying greater deficiency gaps, and the availability and specificity of a firm's resources facilitate the implementation of these 'risky' practices. We thus follow Oliver's insight (1991, 1997) that there might be variance in firms' response to similar institutional stimuli and apply it in the context of environmental innovation.

Scott (1995) identified three basic 'pillars' that structure and provide meaning to organizational behavior: regulative, normative, and cognitive. Regulation provides explicit guidance to organizations through rules, controls, rewards, and sanctions. Norms guide behavior through a less explicit system of standards and values. Cognition includes cultural elements that govern choice often without receiving conscious thought. Although Scott recognized that all institutions combine the three elements, he suggested in a subsequent review of institutional theory (Scott, 2005) that the regulatory and normative dimensions deserve special attention from researchers considering institutional pressures. Although the three elements are often at work simultaneously, they display varying degrees and their relevance is context-specific. In the case of environmental sensitive industries, a significant amount of research has shown that both regulatory agents (e.g., governmental agencies) and normative agents (e.g., professional non-governmental organizations) are relevant actors in shaping the institutional field (Buysse and Verbeke, 2003; Henriques and Sadorsky, 1999; Kassinis and Vafeas, 2006). Next, we turn our attention to the role of these various actors in influencing a firm's environmental innovation efforts.

Regulatory pressures and environmental innovation

Regulatory pressures are typically exerted by governments that ultimately have the power to recognize or deny the organization's existence in their jurisdiction. At the very least, noncompliance with regulations can be very costly to the firm given the government's superior power (Deephouse, 1996). Greater acquiescence could shelter the firm from political risks (e.g., the possibility of close monitoring by government entities), and legal coercion (e.g., the passage of more stringent regulations or tighter enforcement of existing regulations), particularly when the potentially offensive behaviors (such as release of toxic chemicals) involve high social costs.

Pressures from regulators are typically oriented toward environmental improvements in terms of pollution emissions. By definition, environmental

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innovation is geared toward this goal, thereby aligning this type of investment with external demands. Compared with other environmental practices, which might be effective in the short term in signaling the right direction but which are subject to being decoupled from actual implementation (i.e., they might simply be cosmetic), environmental innovation is a proactive initiative that has a longer horizon, requires greater commitment, has superior chances of having bigger a impact in terms of polluting emissions and is thus more likely to reap the social benefits of acquiescence.

While environmental innovation may help a firm reduce or eliminate the toxic burden of production processes to the natural environment, this research may be costly, with uncertain financial returns (Markman, Espina, and Phan, 2004). However, environmental innovation not only helps firms to improve their legitimacy but also to avoid the penalties for poor compliance with regulations (for instance, closer government supervision or protracted litigation). Notice that, although statutory law is important, pressures mostly emanate from its enforcement. Companies have greater incentives to comply when they face both a higher probability to be detected in an illegal behavior (higher monitoring) and/or they are subject to tougher sanctions. This leads to our first hypothesis:

Hypothesis 1: As environmental regulatory pressures increase, environmental innovation will increase accordingly.

This proposition is reminiscent of the so-called Porter's hypothesis, which argues that stricter environmental regulation might lead to innovation and thus competitive advantage (Porter and van der Linde, 1995). We reach this conclusion from a rather different logic, and this is crucial for understanding the contribution of this paper. By envisioning innovation as a means to achieve legitimacy, we can warp institutional logic to explain heterogeneity across firms (see Hypothesis 3). By engaging in environmental innovation, a focal firm can reduce the legitimacy gap with its peers, independently from financial considerations. This logic elucidates the sources of firm heterogeneity in environmental innovation, which are unexplained in Porter and van der Linde (1995). Moreover, while these authors stress the role of regulatory design, we focus on regulatory enforcement as the prominent mechanism in generating pressures upon companies.

Normative pressures and environmental innovation

Normative influences typically come from professional organizations and other focal social actors, which define appropriate behavior and standards for group members (Scott, 2005). These norms, often implicit, are related to the issue of legitimacy. In their quest for legitimacy, organizations compare themselves with their peers and try to behave in accordance with standards or norms prevalent among members that share the same institutional field.

As far as it concerns environmental standards and norms, non-governmental organizations (NGOs) play a key role. Researchers have shown that, although NGOs are outside stakeholders, not necessarily part of formal channels, they can influence organizational processes (King, 2008; Lounsbury, Ventresca, and Hirsch, 2003; Weber, Rao, and Thomas, 2009). Indeed, environmental NGOs not only contribute to creating voluntary standards that push firms beyond minimum regulatory requirements (e.g., the Global Reporting Initiative Guidelines) but can also mobilize resources, exercise their voice, channel and coordinate activist groups and social movement initiatives (Reid and Toffel, 2009). Thus, the presence and activities of environmental NGOs around a company's facilities are likely to increase the pressures that the firm feels to comply with those standards that are considered as being in the same 'ball park,' particularly when strong externalities are involved (as would be the case with pollution).

Several arguments support the notion that environmental NGOs can spur firms' environmental innovation. First, to the extent that environmental NGOs are keenly interested in environmental performance, it seems logical that they would applaud a firm's innovation efforts that are likely to have a salutary impact on that performance. Environmental innovation represents an important behavioral indicator that the firm is trying to meet or exceed environmental performance standards concerning the invention of new designs, and the creation of novel products and processes to reduce or eliminate the use and generation of hazardous substances.

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Second, because environmental NGOs are usually very knowledgeable about environmental issues, they are unlikely to be satisfied with less tangible or ceremonial measures and thus firms cannot hope to gain legitimacy by engaging in more symbolic endeavors. While off-the-shelf alternatives oriented toward meeting minimal environmental standards can be obtained in the open market (Berrone and Gomez-Mejia, 2009), these may not always be suitable for the specific products/production systems of the company. Moreover, they are likely to be considered as simple window dressing by hypercritical stakeholders like environmental NGOs. In this regard, environmental innovation signals a long-term commitment to reduce emissions and offers a more substantive response to institutional demands.

Third, environmental innovation requires an investment with the potential of becoming a source of competitive advantage (unlike off-the-shelf solutions). NGOs can provide the knowledge needed to generate unique and disruptive ideas (Martin, 2002) and can thus be sources of inspiration for novel solutions and new value-creating strategies (Hart and Sharma, 2004). Although they are outside stakeholders, environmental NGOs can trigger internal tensions and struggles, creating fertile environments for innovation and change (King, 2008). Thus, we predict that:

Hypothesis 2: As environmental normative pressures arising from NGOs increase, environmental innovation will increase accordingly.

Heterogeneity in firm responses to institutional pressures

Are some organizations more likely than others to engage in environment-related innovation given similar institutional pressures? If so, what explains this heterogeneous response? Although firms might be subject to exactly the same institutional pressures because, for example, they are located in the same state and are thus subject to a single set of environmental regulations, their responses may differ because of *ex ante* heterogeneity among them (i.e., they have different starting points). Thus, each company responds to the pressures given its current conditions as a constraint. Specifically, we will focus on two sources of *ex ante* heterogeneity across firms: their deficiency gap and their resource profile. A firm's deficiency gap is the result of a combination of managerial decisions, consciously undertaken given the institutional context, and several other factors unrelated with the firm's environmental stance such as inertia, path-dependency, and bad luck. The deficiency gap affects firms' responses to institutional pressures through at least two channels. First, firms that are seen to cause more environmental damage than their peers are more visible to those external actors who exert pressures on them (the government, activists, the media, NGOs, and the like). The fact that they are underperforming makes these firms 'sitting ducks' or easy targets for critics. Hence, companies that find themselves in that position are more likely to be deemed environmentally illegitimate.

Second, managers in firms whose reputations have suffered from accidents resulting in pollution might be more sensitive to environmental pressures (Delmas and Toffel, 2008). Such companies might also face higher risks of business interruptions and legal costs, since poor environmental performance can damage relationships with regulators and other focal social actors (Chatterji and Toffel, 2010). Thus, firms with larger deficiency gaps will impute greater value to environmental projects (such as new methods for treating air or water emissions, recycling or reusing waste, finding cleaner energy sources, and so on) vis-à-vis standard environmental practices in the hope that innovation will enable compliance and help to appease external monitors. We can thus expect that:

Hypothesis 3: A larger deficiency gap between the focal firm's past environmental performance and industry standards strengthens the relationship between environmental pressures (both regulatory and normative) and environmental innovation.

The second driver of heterogeneity in firms' response to environmental institutional stimuli is variation in firms' resource profiles. Both institutional theory and the RBV approach have highlighted the importance of resources. According to the former, greater legitimacy and status enhance relations with resource-provider stakeholders, facilitating access to scarce, critical inputs relevant for survival (Baum and Oliver, 1992). According to the latter (Barney, Wright, and Ketchen, 2001), a firm outperforms its competitors if it manages its overall resources so that

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they generate capabilities that are rare, not substitutable, valuable, and difficult to imitate. Most importantly, strategy itself is constrained by, and dependent upon, a firm's resource profile (Bourgeois, 1981; Cyert and March, 1963; Pfeffer and Salancik, 1978). Resources give the firm leeway in choosing the best strategy in response to external requirements (Sharfman *et al*., 1988) and, at the same time, are critical determinants of knowledge creation and organizational capabilities (Makadok and Barney, 2001). Therefore, internal resources are likely to moderate the relationship between institutional influences and environmental innovation. Specifically, we propose that the effect of institutional pressures on a firm's environmental innovation is contingent on two internal resource dimensions: slack and specificity.

'Organizational slack is that cushion of actual or potential resources which allows an organization to adapt successfully to internal pressures for adjustment or to external pressures for change in policy, as well as to initiate changes in strategy with respect to the external environment' (Bourgeois, 1981: 30). Higher levels of resources provide the firm with greater flexibility toward, and better understanding of, external influences (Cyert and March, 1963). Slack enhances an organization's adaptability because its repertoire of strategic choices is more abundant, and it can respond faster and more effectively than firms with limited resources. For instance, Smith *et al*. (1991) contended that firms with more ample resources can invest in sophisticated information systems that enhance the understanding of external influences, allowing them to respond in a more timely and effective manner. Thus, resource-abundant firms facing environmental pressures should be more capable of securing the necessary material and human talent to launch environmentally friendly products or processes. In contrast, firms with less resources have a reduced ability to change, and are forced to 'put out fires' rather than engage in longer-term initiatives with ambiguous benefits (such as environmental innovation). When there is little slack, the firm must use its scarce resources to attend to its most immediate and pressing needs; hence management may be tempted to neglect environmental demands or respond to them in a more cosmetic way.

Similarly, resources buffer firms from uncertainty and enable them to take risky initiatives in response to external pressures (Chattopadhyay, Glick, and Huber, 2001). Slack allows polluting firms to experiment generously with risky environmental ideas. Clearly, some resources not available within the company might be obtained from the market. For instance, the lack of internal funds can be mitigated by raising capital or by borrowing money. As environmental innovation is both risky and has a long-term horizon, it is plausible to assume that the marginal cost of leveraging external capital for research and development (R&D) activities is much higher than that of using internal available cash. Thus, we can postulate:

Hypothesis 4a: Greater organizational slack strengthens the relationship between environmental pressures (regulatory and normative) on a focal firm and environmental innovation.

The organizational economics literature emphasizes another key feature of resources: their specificity (Williamson, 1985). Specificity of resources involves investments in durable, specialized assets that cannot be easily redeployed from existing uses and users except at a significant loss of productive value or at a deep discount (Dierickx and Cool, 1989). Organizations that are heavily invested in such assets bear greater risk, as business failure would be catastrophic to owners, managers, and employees. Indeed, the larger the share of firm-specific assets of the overall asset portfolio of a company, the higher the value loss in case of default and asset liquidation (Ziedonis, 2004).

In highly polluting sectors, a bad reputation for poor environmental performance might damage relationships with regulators and other focal social actors. Generally speaking, these firms are more likely to face higher business risk in a variety of forms such as loss of key customers, higher rates of bankruptcy, greater legal costs, and government sanctions (Cohen, 1987; Kassinis and Vafeas, 2009). It seems logical that firms confronting these dangers should be more responsive to institutional pressures, as greater legitimacy might provide some degree of protection from financial distress and asset liquidation. However, as we discussed above, the costs and implications of asset liquidation as a result of financial distress are heterogeneous across firms and depend, among other things, on the extent of asset specificity of the overall asset portfolio of the focal company. Clearly, firm-specific assets (not only those related

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to pollution control) are more likely to suffer in value should the firm be required to liquidate them. Thus, for a given level of institutional pressures, firms with more specific assets will be more inclined to seek environmental innovation in order to deal with those pressures. In other words, for firms heavily invested in firm-specific assets, the risk posed by poor compliance with regulations and norms (overall business risk) is far greater than the circumscribed risk associated with the pursuit of environmental innovation. Together, these arguments suggest that:

Hypothesis 4b: Greater specificity of firm assets strengthens the relationship between environmental pressures (regulatory and normative) on a focal firm and environmental innovation.

METHODS

Sample and data collection

We tested our hypotheses with a large sample of firms belonging to the 20 most polluting sectors according to the U.S. Environmental Protection Agency's (EPA) TRI (Toxic Release Inventory) program. After compiling all the necessary information from three different databases (TRI, COMPUSTAT, CHI's Patent Citation Indicators database), we ended up with an unbalanced panel of 326 U.S. publicly traded companies for the period 1997–2001. The final sample of firms was obtained as follows. First, given that polluting industries can be considered a strong organizational field (Berrone and Gomez-Mejia, 2009; Hoffman, 2001), we identified the 20 most polluting U.S. sectors by ranking industrial sectors as defined by the two-digit Standard Industrial Classification (SIC) codes according to the total amount of toxic emissions for the period under analysis. We used the TRI database to produce the ranking.[1]

We then obtained the list of U.S. firms belonging to these top polluting sectors from the COMPUSTAT database. Once we identified the firms, we obtained data about environmental innovation from Nameroff, Garant, and Albert (2004), who used information from the CHI's Patent Citation Indicators database to identify environmental patents between 1983 and 2001 (further details below). This database, which tracks patent information about firms with more than 40 patents granted in the last five years, represents more than 60 percent of all U.S. patents granted since 1992 and more than 70 percent of those patents not held by private individuals.[2] We used multiple sources to associate environmental patents to parent companies. First, we consulted the Mergent manuals, which compile data regarding subsidiaries. In some cases, however, they did not provide the necessary information for all firms in the sample, so we consulted the COMPUSTAT database. In a few remaining cases (less than 10 percent), we consulted the companies' Web sites.

Information regarding institutional forces came from the Environmental Council of the States (ECOS) and from the National Center for Charitable Statistics (NCCS). Finally, the rest of the variables for our study were obtained from the United States Patent and Trademark Office (USPTO), the (EPA), and COMPUSTAT databases.[3]

Measures

Dependent variable

Environmental innovation. In previous studies, environmental innovation has been commonly measured through questionnaire surveys (Anton *et al*., 2004; Christmann, 2000). Unfortunately, these surveys offer little insight into environmental innovation *per se*, since environmental technologies and processes are rarely specifically addressed. Moreover, responses to these questionnaires may be seriously biased, as respondents tend to present a socially desirable image of themselves or their firms. Other studies have used the adoption of environmental management systems (like

[1] The 20 most polluting U.S. sectors for the period analyzed as defined by the two-digit SIC code are 10, 50, 33, 49, 28, 36, 12, 13, 20, 32, 30, 51, 26, 34, 29, 31, 35, 37, 24, and 27.

[2] The fact that the empirical analysis considers firms with at least eight patents per year on average implies that the results of this study apply to large innovative firms.

[3] In 1997 the EPA added seven industrial sectors to TRI, namely, metal mining, coal mining, electrical utilities that combust coal and/or oil, hazardous waste treatment and disposal facilities, chemical wholesale distributors, petroleum bulk stations and terminals, and solvent recovery services. Given the importance in terms of polluting impact of these industries, we decided to start our analysis in the year 1997. Our sample period ends in 2001 because we used environmental patents from Nameroff *et al*. (2004) who have data from 1983 until 2001.

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ISO 14000) as a proxy for environmental innovation. However, by their very nature, off-the-shelf systems tend to be standardized, with an emphasis on more uniform approaches to deal with environmental issues.

Patents are a much more robust indicator of environmental innovation, for several reasons (Kemp, 2008). First, patents are by definition granted only for novel and nonobvious inventions. Second, the exclusive rights that they bestow provide a potential edge over rivals. Third, patent files contain a description of the invention that allows researchers to classify patents according to technological sector, type of use, area of origin, and technical characteristics. Furthermore, patents can be quantified and are publicly available for long time series.

We obtained environmental patents from Nameroff *et al.* (2004), who searched within the CHI's Patent Citation Indicators database for 'green' chemistry patents taken out in the United States.[4] They identified 3,235 green chemistry patents, dispersed broadly across sectors and granted between 1983 and 2001. Among other information, Nameroff and colleagues provided company assignees and the number of forward citations for each of these patents. Their environmental patent search strategy combined examiner-assigned patent classifications with keywords and phrases that appeared in utility patent titles, abstracts, and exemplary claims. They relied on keywords and definitions described by Anastas and Warner (1998), areas of green chemistry identified by the Organisation for Economic Co-operation and Development (OECD) Sustainable Chemistry initiative, and guidance from the EPA's Green Chemistry program. The word list Nameroff *et al.* used to perform the search tried to capture concepts like the design of safer chemicals, the development of processes to minimize energy consumption, the use of solvent-free processes and alternative safer solvents, and so on.[5]

We realized that patent counts have some limitations as measures of innovation output and, thus, we tried to deal with these in several ways. First, not all innovations are patented. Survey data (Cohen, Nelson, and Walsh, 2000) show that the effectiveness, and thus the use of patents as protection mechanisms, strongly vary across sectors. We took this issue into account, at least partially, by controlling for time-invariant differences in the likelihood of patenting a given environmental innovation by sector. Second, some patents are very valuable while others are worth almost nothing. To correct for that, we used the total number of citations received by the patents granted each year to a focal firm to measure environmental innovation (Hall, Jaffe, and Trajtenberg, 2005). Formally, let C_{ikt} be the number of citations received by any given (environment-related) patent k, granted to firm i $(i = 1, \ldots, 326)$ in period t $(t = 1997, \ldots, 2001)$. Citations are computed over the next years following the granting date (year fixed effects partially control for truncation problems). Our dependent variable is thus equal to $\sum_k C_{ikt}$.

Explanatory variables and moderators

Regulatory pressures. In order to operationalize regulatory pressures, we used information from the Environmental Council of the States (ECOS) summarized in Brown and Green (2001). Prior research (Kassinis and Vafeas, 2006) has used the ECOS report as a valid source of pressures. This report provides a summary of the state enforcement and compliance activities between 1995 and 1999 with the goal of offering an accurate picture of what states are doing to enforce environmental laws and helping agents comply with them. One of the most recognized methods of conducting surveillance on the regulated entities is the on-site inspection. Different types of inspections are performed to determine the compliance status of an entity. The report provides the total number of inspections by state per year, which is the central variable we used to compute a measure of regulatory pressures.

The ECOS report also provides information on each state regulated universe, that is, on the total number of regulated entities that may be sites,

[4] Green chemistry R&D is defined as 'the invention, design and application of chemical products and processes to reduce or to eliminate the use and generation of hazardous substances' (Anastas and Warner, 1998: 3). Note that green chemistry patents are used extensively in all polluting industries, not only in chemicals. For instance, in our sample, more than half of the green patents belong to sectors other than the chemical sector.

[5] These are two examples of environmental patents: a) U.S. Pat. No 6.074.995 (13 June 2000) assigned to Lubrizol Corp. relates to a method for reducing engine friction, in turn improving fuel mileage; b) U.S. Pat. No 6.171.568 (9 January 2001) assigned to Degussa-Huls AG relates to a method for purifying exhaust air and/or effluents.

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facilities or incidences. A 'site' is a regulated location without a building, such as a well field; a 'facility' is the typical regulated entity, such as a factory; and an 'incident' might be, for example, a spill on a highway. Each of these entities requires an environmental oversight or regulation by the state environmental agency. Given this information on the total number of regulated entities for each state, we computed the ratio of the total number of inspections and the total number of regulated entities (divided by one thousand) to obtain a measure of the state level of regulatory pressures. In summary, we are assuming that in a given year, firms located in a state with a higher number of inspections per thousand of regulated entities face greater regulatory pressures than those located in a state where a lower number of inspections per thousand of regulated entities took place.

Finally, we used information on the location and the number of a firm's subsidiaries from the Orbis database and computed a weighted average of the state-level measure of regulatory pressures described above, taking into account all those states where the focal firm had a subsidiary and using the proportion of subsidiaries the firm had in each of the states as weights. Given the non-normality of the distribution of this variable, we applied a logarithmic transformation before including it in the model.

Normative pressures. We proxied normative pressures by using information on the number of environmental NGOs per year and state available at the National Center for Charitable Statistics (NCCS), which is the national repository of data on the nonprofit sector in the United States. The basic assumption is that the level of normative pressures that a firm faces increases with the number of NGOs in a given year and state. We divided the total number of environmental NGOs by the total number of regulated entities (divided by one thousand) per year at the state level. Again, for those firms with subsidiaries spread across different states, we computed the weighted average taking into account all those states where the focal firm had a subsidiary and using as weights the proportion of subsidiaries the firm had in each of the states. Given the non-normality of the distribution of this variable, we applied a logarithmic transformation before including it in the model.

Deficiency gap. To compute the deficiency gap, we employed the following formula: $Deficiency\ gap_{it} = \log\{[\sum_j (E_{jist} - Mean(E_{st})] + k\}$ where, E_{jist} denotes the emissions weighted by toxicity of facility j, belonging to firm i, in sector s (four-digit SIC code), at year t. $Mean(E_{st})$ denotes the facility average level of emissions weighted by toxicity in sector s, at year t. Finally, k is a constant such that $[\sum_j (E_{jist} - Mean(E_{st})] + k \geq 1$.

Data on firm emissions were extracted from the EPA's TRI program. Use of TRI data is well established in management research on the environmental impact of corporations (Anton *et al.*, 2004; Klassen and Whybark, 1999; Russo and Harrison, 2005). Under the EPA's Emergency-Right-to-Know Provision, industrial facilities that have 10 or more full-time employees and that release any listed toxic substance in excess of minimum reporting thresholds to any of four different media (air, water, land surface, and land subsurface via underground injection) are required to report the type and amount of emissions to the EPA (EPA, 2002). TRI data reflect releases of noxious emissions, but not the level of exposure to the public or the ecosystem (EPA, 2002). While recent studies have tried to mitigate this shortcoming by weighting emissions by the toxicity level of chemical agents (e.g. King and Lenox, 2002; Russo and Harrison, 2005), these weighting systems have their limitations as they do not account for the medium (e.g., air, water, land) into which the chemical agent is released. To address this drawback, we weighted chemicals using the Human Toxicity Potential (HTP) factor developed by Hertwich and colleagues (2001) in a given year considering the medium into which it was released (Berrone and Gomez-Mejia, 2009). This allows for a more precise assessment of the threat that chemical releases pose to human health. We then aggregated the results first across chemicals and next at facility level. Once we had this facility-level measure of emissions, we subtracted its corresponding industry average at the four-digit SIC code for every facility-year observation, so positive values of this variable indicate higher emissions with respect to a facility's corresponding industry and vice versa for negative values.[6] Then we averaged facility-level deviations by parent

[6] Previous studies (e.g., King and Lenox, 2000) constructed similar measures of facility excess releases compared to their industry. These works focused on changes in the environmental efficiency of a firm's production processes, and thus have taken facility or firm size into account. We are interested in the potential damage in terms of environmental reputation due to the deficiency gap. This is likely to be associated with excess

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company and obtained an annual observation per company (our unit of analysis). We aggregated facility-level data using the Data Universal Numbering System (DUNS) at the parent level.[7] We rescaled this variable by adding a constant so that its minimum value became 1 and then calculated the logarithm to reach normality. To test for the moderating role of the deficiency gap (Hypothesis 3) we interacted this variable with our measures of regulatory and normative pressures.

Slack resources. We included two different measures of organizational slack based on previous literature (Bromiley, 1991; Fleming and Bromiley, 2003): (a) the ratio of working capital to sales, and (b) the ratio between net income and revenues (profit margin). Both variables were obtained from COMPUSTAT. Although their correlation is low, suggesting that they might capture different dimensions of organizational slack, they behave remarkably similarly in the regressions. Given the non-normality of these two variables, we applied the logarithm. To test for the moderating role of organizational slack (Hypothesis 4a), we interacted these two variables with our measures of regulatory and normative pressures.

Asset specificity. Researchers have approached the specificity of firm assets with a number of different proxies (David and Han, 2004). Lacking a direct measure of asset specificity (usually obtained through questionnaires), we relied on secondary data to proxy this construct. Specifically, we followed Ziedonis (2004) and used the logarithm of the ratio of the book value of a firm's machinery and equipment to the number of employees (from COMPUSTAT). Investments in production plants, and machinery and equipment are typically large and these assets are usually not easy to redeploy. They are also less likely to be valuable in other firms should the company fail and its assets be liquidated. For example, a chemical company that invests in a plant for producing polypropylene in a given geographical market will find it difficult to recover most of its initial investment should the asset be sold out. This variable was interacted with our measures of regulatory and normative pressures to test Hypothesis 4b.

Control variables

R&D intensity. Since R&D intensity is a central input of the patent development process, we included it as a control variable in the estimated models. We used the total amount of R&D expenditures divided by the total number of employees (from COMPUSTAT), to avoid confounding the effect of R&D investment with a size effect. Finally, given the non-normality of the distribution of R&D investment, we took the logarithm so that our final measure of R&D intensity is the logarithm of R&D expenditures per employee. A dummy variable was used to control for those firms that did not report, or had missing information about R&D expenditures.

Number of patents. We included the total number of patents granted at the USPTO as a control variable for four reasons. First, this variable partially controls for a firm's ability to develop innovations (DeCarolis and Deeds, 1999). Second, previous research suggests that the number of patents reflects a firm's technology portfolio (Ahuja and Lampert, 2001; Meyer, 2002). Third, the number of patents controls for factors that affect a firm's propensity to patent, such as top executives endorsing an aggressive patenting policy. Lastly, although we include sector dummies that also control for the differences in patenting incentives across sectors, the total number of patents captures time-varying firm-level differences (within sectors), which have also been reported in the empirical literature on patents (Cohen *et al*., 2000; Levin *et al*., 1987).

Firm size. We controlled for potential economies of scale in patenting by including the variable firm

releases in absolute terms rather than per unit of production. Thus, our measure does not account for facility or firm size.

[7] On occasions, DUNS numbers were missing or incorrect. To account for this issue, we engaged in a painstaking and tedious process that involved, as a first step, the manual matching of publicly traded companies included in both the TRI and COMPUSTAT databases. In order to identify the corresponding DUNS numbers of these firms, we used the most frequent DUNS number reported by facilities indicating that they belonged to these firms. We cross-checked this data over all years in our sample to make sure that the DUNS number assigned to a parent company was correct. To identify plants that belonged to these firms, we cross-checked several fields related to the facility's address across years to see if a given facility reported the parent's DUNS in a subsequent year and used it when available. If the DUNS number was not included during our entire time span, the facility was not considered. When address fields were identical but the parent's DUNS were not, we checked whether the DUNS number was completely different or was simply incomplete or misreported. We considered a DUNS number incomplete or misreported if the first (or the last) four digits of the reported number were identical to that reported in another year. When incomplete, they were completed and when different they were discarded.

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size, measured as the logarithm of the number of employees (from COMPUSTAT).

Miscellaneous controls. All the estimated models include sector (from COMPUSTAT), and year fixed effects to control for unobserved heterogeneity between sectors (at the two-digit level SIC codes) and for annual trends that could affect the level of environmental innovation. All interacting variables are included in the regressions as controls for a correct interpretation of the coefficients. Continuous variables are standardized to facilitate the interpretation of the results.

Empirical model and estimation method

Given the count nature of the dependent variable (i.e., number of citations), we needed to use nonlinear estimators such as Poisson and negative binomial regression models. Preliminary exploratory analysis of the data using the Lagrange Multiplier test rejected the pure Poisson model in favor of a model where the variance is proportional to the mean. Thus, for the purpose of this paper, we present results only for the negative binomial estimations.

Given the nature of the innovation process, its serendipity and complexity, it is quite unlikely that the explanatory variables (most importantly, institutional pressures) immediately generate environment-related patents. For this reason, we explored different lag structures in the empirical analysis. Results below are reported with a two-year lag for all explanatory variables that are time variant.

RESULTS

Descriptive statistics and correlations are reported in Table 1. The untransformed value for firm size averages 32,161 employees, indicating that our sample is composed of very large firms and hence their overall environmental impact is likely to be substantial. The untransformed values for regulatory pressures is 519 and for normative pressures is 58, reflecting the average number of inspections and the average number of environmental NGOs for every thousand of regulated sites, respectively.

Results of the negative binomial estimations are presented in Tables 2 and 3. As a summary measure of the magnitude of the estimated effects, we also report the average marginal effects (in brackets). All specifications in Tables 2 and 3 include the log of the ratio of working capital to sales as the proxy for organizational slack. Using the 'log of the profit margin' yields results that are qualitatively unchanged and are available from the authors upon request. The estimation of the basic specification, which includes only control variables, is shown in Model 1. As we expected, firms with more patents, bigger firms, firms with more emissions with respect to their sector, firms with higher R&D intensity, and firms with more specific assets generate more environmental innovation.

We first examine the main effect of institutional pressures on environmental innovation (Hypotheses 1 and 2) by including our two-year lagged measures of institutional forces in Model 2 (Table 2). Both regulatory (Hypothesis 1) and normative (Hypothesis 2) pressures have a positive and significant effect on the level of a firm's environmental innovation, providing support for both hypotheses. More precisely, a change of one standard deviation in regulatory pressures (normative pressures) increases environmental innovation by 1.49 (1.14). Given our measures of regulatory and normative pressures and environmental innovation, these estimated average marginal effects imply that 100 additional inspections (per thousand of regulated sites) are associated with almost 0.5 additional citations (to a firm's environmental patents), while 10 additional NGOs (per thousand of regulated sites) are associated with 0.35 additional citations (to a firm's environmental patents). To put these figures in perspective, notice that the sample mean for inspections is 519, for NGOs is 58, and for citations to environmental patents is 0.42.

In Models 3, 4, and 5 (Table 3), we include the interactions between both types of institutional pressures and the deficiency gap (Hypothesis 3), organizational slack (Hypothesis 4a) and asset specificity (Hypothesis 4b), respectively.

In Model 3 we find that the average marginal effect of the interaction terms are positive and highly significant for both regulatory (1.22) and normative pressures (1.87), providing evidence that the deficiency gap positively moderates the effect of institutional pressures. For a given level of institutional pressures, firms with a larger deficiency gap with respect to its industry standards will develop more environmental innovation.

In Model 4 of Table 3, we explore the moderating effect of the availability of resources on the

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Table 1a. Descriptive statistics

	Mean	St. dev.	Min.	Max.
Environmental innovation	0.42	2.38	0.00	45
Firm size (log-employees)	2.37	1.52	0.27	6.47
Total patents	53.68	162.13	0.00	1643
R&D intensity	0.16	0.45	0.04	5.47
Deficiency gap	12.35	9.90	0.00	24.92
Org. slack (1)	−1.48	1.57	−4.76	5.54
Org. slack (2)	1.76	1.86	−1.73	4.15
Asset specificity	2.57	0.80	0.03	8.23
Regulatory pressures	5.81	0.98	1.49	7.91
Normative pressures	3.79	0.83	0.30	4.79
No R&D (dummy)	0.08	0.27	0.00	1

Table 1b. Correlations

	1.	**2.**	**3.**	**4.**	**5.**	**6.**	**7.**	**8.**	**9.**	**10.**
1. Environmental innovation	1									
2. Firm size (log-employees)	0.08	1								
3. Total patents	0.05	0.30	1							
4. R&D intensity	0.02	−0.33	−0.05	1						
5. Deficiency gap	0.13	0.42	0.16	−0.23	1					
6. Org. slack (1)	−0.01	−0.08	−0.03	−0.53	0.08	1				
7. Org. slack (2)	0.01	−0.10	0.02	−0.51	0.07	0.08	1			
8. Asset specificity	0.11	0.27	0.18	−0.09	−0.12	−0.09	−0.1	1		
9. Regulatory pressures	0.05	−0.08	−0.10	−0.0	−0.24	0.17	−0.17	−0.11	1	
10. Normative pressures	0.04	−0.07	−0.04	−0.14	−0.08	0.23	−0.23	−0.06	0.65	1
11. No R&D (dummy)	−0.03	−0.10	−0.09	−0.10	−0.08	−0.03	−0.03	−0.18	−0.04	−0.04

Org. slack (1) is the log of the ratio of working capital to sales; Org. slack (2) is the log of the ratio between net income and revenues (profit margin).

generation of environmental innovation (Hypothesis 4a). While the average marginal effect of organizational slack is positive and significant for normative pressures (0.71), it is negative and significant for regulatory pressures (−1.59). In other words, at higher levels of slack, the positive effect of normative pressures on environmental innovation becomes stronger, but the opposite seems to be true for regulatory pressures. This result provides mixed support for Hypothesis 4a. We discuss plausible explanations for this finding in the following section.

Next, in Model 5 of Table 3, we explore the extent to which the specificity of the firm assets moderates the effect of institutional pressures on environmental innovation (Hypothesis 4b). We find, as suggested by Hypothesis 4b, that both regulatory and normative pressures induce more environmental innovation in firms with higher levels of specific assets (average marginal effects of 3.42 and 2.71, respectively).

Note that, as we explain in the robustness tests section, the sign of the average marginal effects described above for each of the interaction terms are consistent across different combinations of values of the interacted variables. Finally, Model 6 includes all possible interaction terms. One plausible concern with this specification is the existence of multicolinearity. We explore this possibility by computing the variance inflation factors (VIFs) for each linear regression coefficient of the models reported in Table 3. While for the models that test one moderator effect at a time (Models 3, 4, and 5) the VIFs are always smaller than 3, in Model 6, some of the VIFs are near 10, suggesting that colinearity becomes an issue in this specification. However, although some standard errors may be inflated, significantly reducing the precision of the estimates, most of the significant effects reported in the first three models of Table 3 remain significant in Model 6. In addition, we do not observe shifts in the sign of the estimated coefficients

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Table 2. Negative binomial estimations of the determinants of environmental innovation

	Dependent variable: environmental innovation	
	(1)	(2)
Firm size$_{(t-2)}$	0.89***	0.73***
	(0.39)	(0.43)
	[1.78]	[1.65]
Total patents$_{(t-2)}$	0.27*	0.57**
	(0.40)	(0.48)
	[0.78]	[1.12]
R&D intensity$_{(t-2)}$	1.64***	1.43***
	(0.52)	(0.37)
	[1.41]	[1.26]
Deficiency gap$_{(t-2)}$	0.88***	0.75***
	(0.45)	(0.86)
	[4.21]	[3.22]
Organizational slack$_{(t-2)}$ (working capital/sales)	−1.70	−0.96
	(1.11)	(0.49)
	[-0.95]	[-0.54]
Asset specificity$_{(t-2)}$	1.28***	1.37***
	(0.41)	(0.38)
	[1.18]	[1.34]
Regulatory pressures$_{(t-2)}$		1.49***
		(0.88)
		[2.31]
Normative pressures$_{(t-2)}$		1.14**
		(0.81)
		[1.65]
No R&D$_{(t-2)}$	0.30	0.25
	(0.45)	(0.64)
	[0.27]	[0.23]
Observations	739	739
Log likelihood	−157.44	−152.39

Reported coefficients are standardized. Average marginal effects are reported in brackets. Standard errors are reported in parentheses and computed using the Delta method. All models include a constant, industry and year fixed effects. Significance levels:
* $p < 0.05$,
** $p < 0.01$, and
*** $p < 0.001$.

when introducing changes in the specification. All these elements suggest that, although some degree of colinearity may exist in the last specification (as is common in cases where multiple interactions are tested), results remain stable.

To gain further insights, in addition to the marginal effects reported in Table 3, we have also computed the average marginal effects of both regulatory and normative pressures for two specific values (the 5th and the 95th percentiles) of each of the moderating variables using Models 3, 4, and 5, respectively. Results from Model 3 indicate that, for the 5th percentile (95th percentile) of the deficiency gap, 100 additional inspections are associated with 0.53 (0.79) additional citations to environmental patents, while 10 additional NGOs are associated with 0.05 (0.20) additional citations. Results from Model 4 indicate that, for the 5th percentile (95th percentile) of organizational slack, 100 additional inspections are associated with 0.20 (0.04) additional citations to environmental patents, while 10 additional NGOs are associated with 0.01 (1.88) additional citations. Finally, results from Model 5 indicate that, for the 5th percentile (95th percentile) of capital intensity, 100 additional inspections are associated with 0.34 (0.42) additional citations to environmental patents, while 10 additional NGOs are associated with 0.05 (0.15) more citations.

Robustness tests

To assess the robustness of our findings, we ran several other estimations that we describe next. These robustness tests are available from the authors upon request. We verified that the reported effects are robust to the use of different measures of the dependent variable. Results held unchanged using environmental patent counts instead of citation counts, the sum of citations plus patents, or other value weighted measures previously employed in the empirical literature on patents like, for instance, weighting each patent by the proportion of the citations received by the firm with respect to the number of citations received by firms from the same sector. We also verified that results held qualitatively unchanged when using our alternative measure of organizational slack, the log of the profit margin. Another concern with the baseline specification was that the interaction terms might be picking up time-varying industry effects. To address this concern, we augmented the baseline specification with industry-year fixed effects. The basic results were not affected. We also included the interaction between the no R&D dummy variable and our measures of institutional pressures, in case the effect of institutional pressures was different for these particular firms compared to the rest of the sample. The marginal effects of these interaction terms were not significant, and the introduction of these controls did not change the overall results. As we discussed above, we also experimented with different lag structures for our explanatory

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Table 3. Negative binomial estimations of the determinants of environmental innovation: moderator effects

	Dependent variable: environmental innovation			
	(3)	(4)	(5)	(6)
Firm size$_{(t-2)}$	0.95***	0.53*	0.76**	0.87***
	(0.32)	(0.23)	0.26	(0.27)
	[0.80]	[0.43]	[0.59]	[0.65]
Total patents$_{(t-2)}$	0.63*	0.50**	0.61**	0.61**
	(0.27)	(0.20)	(0.22)	(0.23)
	[0.53]	[0.44]	[0.47]	[0.58]
R&D intensity$_{(t-2)}$	1.25***	1.05**	0.93*	0.95**
	(0.39)	(0.37)	(0.37)	(0.35)
	[1.06]	[0.82]	[0.73]	[0.79]
Deficiency gap$_{(t-2)}$	0.61**	0.62***	0.55**	0.57**
	(2.42)	(0.71)	(0.92)	(0.80)
	[5.2]	[2.12]	[1.86]	[1.74]
Organizational slack$_{(t-2)}$ (working capital/sales)	−0.75	−0.84	−0.48	−0.96
	(0.61)	(0.21)	(0.49)	(1.01)
	[-0.59]	[-0.15]	[-0.18]	[-0.68]
Specific assets$_{(t-2)}$	1.14***	1.11***	4.74***	4.26***
	(0.32)	(0.22)	(0.61)	(0.51)
	[0.97]	[0.73]	[1.85]	[1.72]
Regulatory pressures$_{(t-2)}$	4.04***	0.78***	4.94***	7.82***
	(1.31)	(0.25)	(0.69)	(1.42)
	[3.43]	[0.97]	[1.93]	[4.54]
Normative pressures$_{(t-2)}$	0.85	0.54	0.88	0.17
	(1.26)	(0.32)	(0.65)	(0.86)
	[0.72]	[0.33]	[0.49]	[0.13]
Deficiency gap$_{(t-2)}$*regulatory pressures$_{(t-2)}$	0.09***			0.04*
	(0.46)			(0.22)
	[1.22]			[0.43]
Deficiency gap$_{(t-2)}$*normative pressures$_{(t-2)}$	0.48***			0.08*
	(0.52)			(0.25)
	[1.87]			[0.52]
Org. slack$_{(t-2)}$*regulatory pressures$_{(t-2)}$		−0.14**		−0.07
		(0.71)		(0.43)
		[-1.59]		[-0.41]
Org. slack$_{(t-2)}$*normative pressures$_{(t-2)}$		0.32**		0.06
		(0.32)		(0.37)
		[0.71]		[0.21]
Specific assets$_{(t-2)}$*regulatory pressures$_{(t-2)}$			0.39***	0.59*
			(1.13)	(1.23)
			[3.42]	[2.43]
Specific assets$_{(t-2)}$*normative pressures$_{(t-2)}$			0.49***	0.42*
			(1.12)	(0.92)
			[2.71]	[1.89]
No R&D$_{(t-2)}$	0.25	0.08	0.11	0.29
	(0.54)	(0.43)	(0.53)	(0.62)
	[0.21]	[0.01]	[0.04]	[0.17]
Observations	739	739	739	739
Log likelihood	−148.50	−150.37	−151.47	−147.98

Reported coefficients are standardized. Average marginal effects are reported in brackets. Standard errors are reported in parentheses and computed using the Delta method. All models include a constant, industry and year fixed effects. Significance levels:
* $p < 0.05$,
** $p < 0.01$, and
*** $p < 0.001$.

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variables. Reassuringly, a one-year lag or a three-year lag provided similar qualitative results (although, in general, coefficients became less significant with these other lag structures).

Interpreting and summarizing an interaction term in nonlinear models is difficult since its magnitude and its sign can differ across observations (Hoetker, 2007; Huang and Shields, 2000). Therefore, in addition to the average marginal effects reported in Table 2, we computed marginal effects for different values of the interacted variables to explore possible changes in sign and magnitude. Results were confirmed across different combinations of values of the interacted variables. A graphical representation of the results is available from the authors upon request.

Finally, we explored some possible endogeneity concerns arising from the use of green chemistry patents as a proxy for environmental innovation. First, there may be measurement error in the dependent variable resulting from the existence of other green but non-chemical patents that we are not accounting for with our current measure of environmental innovation, but may be correlated with the strength of institutional pressures. A second possible source of estimation bias may come from the existence of other unobservable factors (apart from the measurement error) that could also be correlated with the level of institutional pressures and the number of green chemistry patents. As both sources of endogeneity described above appear to be more likely within chemical firms, we reestimated all models in Table 3 excluding chemical firms from the sample. Overall, results are qualitatively unchanged with the restricted sample suggesting that focusing on green chemistry patents as a measure of environmental innovation is not producing a bias and that we can trust the estimated effects reported in Table 3.

DISCUSSION AND CONCLUSION

In this study, we proposed that regulatory and normative pressures concerning environmental issues spawn innovative responses in firms and that this link is contingent on a firm's past environmental performance relative to its industry peers, and the availability and type of internal firm resources. Our results have important implications for research and practice.

Contributions and implications for research

Our study reveals important theoretical and empirical contributions to and implications for different streams of management research. Theoretically, we contribute to the institutional, innovation, and RBV literatures, linking domains that generally coexist as silos and using this hybrid approach to better understand why some firms are more likely to pursue environmental innovations than others. We have argued that institutional pressures may actually prompt heterogeneity rather than isomorphism in polluting industries as participating organizations engage in a 'problemistic search' to discover new environmental methods that are idiosyncratic or proprietary to the focal firm. Presumably, these internally generated innovations should enable firms subject to strong institutional pressures to reduce offensive actions (such as toxic emissions) and, thus, through unique approaches (e.g., in the handling and disposal of dangerous material) better comply with the desires of external actors. When institutional pressures are weak, firms should be less inclined to seek differentiation through an active program of environmental innovation, as these risky investments may not be justified by legitimacy gains.

Our framework could also affect the way researchers examine competitive advantage creation (cf. Porter and Van der Linde, 1995), as it suggests that institutional pressures may come into play when resources and capabilities are developed. Specifically, institutional pressures may have an impact on technological change within the firm since they can help refocus current R&D efforts toward environment-related projects. Thus, institutional pressures can influence internal resource allocations and thus be a source of competitive advantage, as they establish the boundaries within which organizations can be creative and inventive. We recognize that we do not explore how this internal process actually unfolds. More fine-grained research is still needed to fully elucidate how these processes work at the individual level.

Extant research has shown empirically that firms' responses to institutional pressures are affected by organizational characteristics including ownership structure, board of director interlocks, and geographic proximity to peer organizations. We extend this line of research by showing that both regulatory and normative pressures can have differential effects on the focal firm's

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environmental innovation that are contingent on its deficiency gap and its resource profile. First, we have shown that firms that report more environmental damage than their industry peers are more prone to engage in environmental innovation compared to those firms that match or surpass the industry's environmental standard. Because poor performers are in a position of legitimacy disadvantage, they are likely to place greater value on environmental innovation as a valid vehicle to boldly respond to environmental concerns and restore their impaired legitimacy. Second, we found that firms with larger investments in specific assets are more likely to respond to institutional pressures by engaging in more environmental innovation. Because these firms are more at risk than other firms with lower levels of specific assets, they are more likely to use environmental innovation as a strategic shield to safeguard their assets from potential losses (King and Lenox, 2000). This prediction is consistent with behavioral agency theory (Wiseman and Gomez-Mejia, 1998).

Third, and contrary to our expectation, we found that the moderator effect of organizational slack differs according to the type of institutional pressure. While normative pressures have a larger positive impact on environmental innovation in firms with more organizational slack, regulatory pressures have a smaller effect in those firms. These results are interesting and challenging to interpret. One would expect that government regulators that enjoy strong enforcing mechanisms (like economic penalties or even the power to shut down a facility) with devastating (and uncertain) effects would generate greater pressures than normative institutions that may not have such strong coercive mechanisms and where the cost of not complying may come in the form of legitimacy sanctions. One possible explanation for these results is that firms with sufficient resources may prefer to pay legal and economic sanctions rather than change products or processes by engaging in risky environmental projects, and use any slack as a 'sandbag' rather than a resource for innovation. It may also be that social sanctions and legitimacy threats emanating from normative actors have even greater power than the economic penalties that regulatory bodies can impose. A third explanation may be that regulatory pressures can easily be satisfied with symbolic or reluctant efforts while normative pressures need more fundamental responses, as the people or organizations exerting these pressures are usually very knowledgeable about environmental issues. Together, these alternative explanations suggest that firms use their slack contingent on the type and strength of the pressures they face, using slack to resist when they face regulatory pressures and to acquiesce to social demands when they face normative pressures.

This paper also fuels the ongoing debate regarding the effectiveness of government in promoting environmentally desirable outcomes. Our results are consistent with previous work that highlights the importance of regulatory forces in spawning corporate social action (King and Lenox, 2000; Newton and Harte, 1997). However, pressures from regulations do not seem to be effective in spurring environmental innovation for firms with high levels of organizational slack. In any case, our results should be viewed with caution as our measure of regulatory pressures does not distinguish the type or duration of enforcement actions. Further analysis of environmental regulations and their enforcement mechanisms could shed some light on how organizational slack is used when firms face different types of regulatory pressures.

Implications for managers

External pressures for more corporate responsibility require environmental strategies to conform to institutional demands, ensure societal legitimacy, and secure organizational success. Given that a firm's competitive advantage depends on its ability to innovate in ways that its rivals cannot easily imitate, environmental innovation appears to be a valuable policy for managers to follow. Our work suggests that external influences may inspire managers to develop unique resources and, indeed, that managers in firms whose reputations have suffered from environmental misconduct or compliance problems might be more sensitive to pressure from regulators and institutions concerned with environmental issues than managers in other companies. However, this does not imply that managers need to wait for criticism before developing environmental innovation. They can take the lead on this type of activity since it should have a salutary effect on firm performance. Finally, managers should understand that technology responses are not a simple reaction to external pressures; there is also the matter of resource specificity—managers should turn their attention to the efficient use of their resources.

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